PING WANG

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RESEARCH INTERESTS

- Computational Imaging & Photography
- Computer Vision
- Image & Video Processing

EMPLOYMENT

Westlake University — Hangzhou, China
Postdoctoral Researcher in Computational Imaging and Machine Vision — *Jul. 2025 – Present*

EDUCATION

Zhejiang University — Hangzhou, China
*Joint Ph.D. Program with **Westlake University** in Computer Science and Technology* — *Sep. 2021 – Jun. 2025*
- Thesis: "Deep Learning for Compressive Imaging Inverse Problems"
- Advisor: Prof. Xin Yuan

Southwest University — Chongqing, China
M.Eng. in Electronic and Communication Engineering — *Sep. 2017 – Jun. 2020*
- Thesis: "Research on Data Privacy Preservation Based on Compressive Sensing"
- Advisor: Prof. Xin He Co-advisor: Prof. Yushu Zhang

Wuhan Institute of Technology — Wuhan, China
B.Sc. in Opto-Electronics Information Science and Engineering — *Sep. 2013 – Jun. 2017*

PAPERS

* co-first author, † corresponding author

1. Z. Shi, L. Wang, Y. Zhong, **P. Wang**†, X. Yuan†, "ShiftSCI: Efficient Hyperspectral Image Reconstruction with Dispersion-Guided Shifting," *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.
2. Z. Shi, L. Wang, X. Wang, Y. Zhong, **P. Wang**†, X. Yuan†, "Physically-Coupled PAN-Guided Unfolding for Dual-Camera Compressive Hyperspectral Imaging," *ACM International Conference on Multimedia (ACM MM)*, 2026.
3. Y. Luo, C. Wu, C. Li, **P. Wang**, X. Yuan, K. Yang, J. Zhang†, "UP-Dem: Deep Unrolling Convolutional Sparse Coding for Color Polarization Image Demosaicking," *IEEE Transactions on Image Processing*, 2026.
4. Y. Song*, Y. Luo*, **P. Wang**†, X. Jiang, X. Yuan†, "Pol-CACTI: A System and Dataset for High-Speed Polarized Video Compressive Imaging," *European Conference on Computer Vision (ECCV)*, 2026.
5. Y. Luo*, C. Li*, Y. Song, **P. Wang**, B. Shi, J. Zhang†, X. Yuan†, "PolarAPP: Beyond Polarization Demosaicking for Polarimetric Applications," *European Conference on Computer Vision (ECCV)*, 2026.
6. Z. He, **P. Wang**†, X. Wang, C. Cheng, and X. Yuan†, "Joint Spectral Image Reconstruction and Semantic Segmentation with Cooperative Unfolding," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
7. X. Jiang, L. Wang, **P. Wang**, and X. Yuan†, "DetectSCI: Toward Object-Guided ROI Reconstruction for High-Resolution Video Snapshot Compressive Imaging," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
8. W. Wan, **P. Wang**, H. Liu, H. Hu, Q. Liu†, "Progress in Computational Optical Imaging Based on Generative Artificial Intelligence," *Acta Photonica Sinica*, 2025.
9. G. Qu, **P. Wang**†, S. Zheng, and X. Yuan†, "Breaking Measurement Barriers: From Compressed Sensing to Deep Reconstruction," *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
10. H. Zhang, M. Cao, X. Yu, H. Luo, Y. Piao, M. Qin, Z. Li, **P. Wang**, X. Yuan†, "High-Speed FHD Full-Color Video Computer-Generated Holography," *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

11. X. Wang, Z. He, **P. Wang**, L. Wang, Y. Hu, and X. Yuan†, "Spectral Compressive Imaging via Chromaticity-Intensity Decomposition," *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
12. Y. Zhang, X. Liu, L. Wang, **P. Wang**, G. Yuan, M. Ku Chen, K. Jiang, X. Yuan†, Z. Geng†, "High-speed video imaging via multiplexed temporal gradient snapshot," *Advanced Photonics Nexus*, 2025.
13. B. Shi, S. Qi, **P. Wang**, and Y. Zhang†, "Hard exif: Protecting Authorship Through Image Metadata, Content, and Hardware," *IEEE Transactions on Image Processing*, 2025.
14. **P. Wang***, L. Wang*, G. Qu, X. Wang, Y. Zhang, and X. Yuan†, "Proximal Algorithm Unrolling: Flexible and Efficient Reconstruction Networks for Single-Pixel Imaging," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
15. **P. Wang**, Y. Zhang, L. Wang, and X. Yuan†, "Hierarchical Separable Video Transformer for Snapshot Compressive Imaging," *European Conference on Computer Vision (ECCV)*, 2024.
16. G. Qu*, **P. Wang***, and X. Yuan†, "Dual-Scale Transformer for Large-Scale Single-Pixel Imaging," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
17. **P. Wang** and X. Yuan†, "SAUNet: Spatial-Attention Unfolding Network for Image Compressive Sensing," *ACM International Conference on Multimedia (ACM MM)*, 2023.
18. **P. Wang**, L. Wang, and X. Yuan†, "Deep Optics for Video Snapshot Compressive Imaging," *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
19. H. Zhang, **P. Wang**, and X. Yuan†, "Deep Learning for Snapshot Image Compressive Sensing," *Imaging Systems and Applications*, 2023.
20. **P. Wang**, L. Wang, M. Qiao, and X. Yuan†, "Full-Resolution and Full-Dynamic-Range Coded Aperture Compressive Temporal Imaging," *Optics Letters*, 2023.
21. Y. Zhang, **P. Wang**, et al, "Privacy-Assured FogCS: Chaotic Compressive Sensing for Secure Industrial Big Image Data Processing in Fog Computing," *IEEE Transactions on Industrial Informatics*, 2020.
22. Y. Zhang, **P. Wang**, et al, "Secure Transmission of Compressed Sampling Data Using Edge Clouds," *IEEE Transactions on Industrial Informatics*, 2020.
23. **P. Wang**, Y. Zhang, X. He, et al, "Big data Privacy Protection Based on Secure Compressive Sensing," *Big Data Research*, 2020.
24. **P. Wang**, X. He, Y. Zhang, et al, "A Robust and Secure Secret Image Sharing Scheme with Personal Identity Information Embedded," *Computers & Security*, 2019.

PATENT

- Y. Zhang, **P. Wang**, L. Y. Zhang, Y. Xie, "A Copyright-Protected and Privacy-Preserving Media Sharing Scheme," AU Patent, Application No. 2018100682, 2018.

RESEARCH EXPERIMENCE

Research Assitant May 2021 – Aug. 2021
School of Engineering, Westlake University *Hangzhou, China*

- Mentor: Prof. Xin Yuan
- Research Topics: Optimization Algorithms for Inverse Problems, Cryptograpgic Compressed Sensing

Research Assitant Sep. 2020 – Apr. 2021
Cyber Science Research Institute, Sichuan University *Chengdu, China*

- Mentor: Prof. Xingshu Chen, Prof. Changxiang Shen
- Research Topics: Computation Outsourcing, Serverless Computing

Research Intern May 2019 – Sep. 2019
Chengdu Industrial Institute, China Mobile *Chengdu, China*

- Research Topics: Image Transmission and Processing for 5G-connected UAVs.

Leadership & Membership

Founder of Westlake Reading Club at Westlake University — Nov. 2021 – Jun. 2023
Student Member of IEEE — Nov. 2021 – Present

Academic Activities

Reviewer

- **Conference**: CVPR 2023-2025, ICCV 2023/2025, NeurIPS 2023-2024, ICLR 2023-2024, ACM MM 2025.
- **Journal**: TPAMI, TIP, Optics Letters, Information Sciences, Pattern Recognition.

Presentations

- "Compressive Imaging Algorithms: from Image to Video." Invited talk at *Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences*. Xi'an, China. Oct. 12, 2024.
- "Deep Models for Inverse Problems in Compressive Imaging." Oral presentation at *4th International Conference on Computational Imaging (CITA 2024)*. Xiamen, China. Sep. 20, 2024.
- "Deep Optics Meets Compressive Imaging." **Best oral presentation** at *3th International Conference on Computational Imaging (CITA 2023)*. Beijing, China. Oct. 30, 2023.

Honors and Awards

- Excellent Graduate Award at Zhejiang University — 2024
- Exploration Award at Westlake University — 2024
- Best Oral Presentation Award at CITA 2023 (10/175) — 2023
- Suwu Scholarship (2.6%), Learning & Innovation Award (3.4%) at Westlake University — 2023
- Merit Graduate Award, Excellent Graduate Award at Zhejiang University — 2023
- Excellent Graduate Cadre Award at Zhejiang University — 2022
- Outstanding Graduate Award at Southwest University — 2020
- Outstanding Academic Innovation Award at Southwest University — 2019
- Excellent Research Award at Southwest University — 2018/2019
- Merit Graduate Award at Southwest University — 2018

Skills and Interests

Language: English – Professional working proficiency (CET-6: 501), Chinese – Native proficiency.
Programming: Python, Pytorch, Matlab.
Interest: Reading, Cooking, Backpacking, etc.

Referees

Dr. Xin Yuan
Associate Professor, College of Engineering, Westlake University
Google Scholar: http://scholar.google.com/citations?user=cS9CbWkAAAAJ&hl
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Dr. Yulun Zhang
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Dr. Yushu Zhang
Professor, College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics
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